Ket ($|\psi\rangle$):

The ket notation represents a column vector in a Hilbert space. It's typically used to represent quantum states. For example, if we have a qubit in the state

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

where α and β are complex numbers representing probability amplitudes, we can write this state using ket notation as

$$|\psi\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

Here, $|0\rangle$ and $|1\rangle$ are the computational basis states of the qubit.

Bra ($\langle\psi|$):

The bra notation represents the conjugate transpose of a ket vector. In other words, it represents a row vector in the Hilbert space, which is the complex conjugate of the corresponding ket vector. For example, if we have the ket vector

$$|\psi\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

then its bra counterpart is written as

$$\langle\psi| = \begin{pmatrix} \alpha^* & \beta^* \end{pmatrix}$$

where α^* and β^* are the complex conjugates of α and β, respectively.

Here's a simple example to illustrate how ket and bra notations work together:

Suppose we have a qubit in the state

$$|\psi\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{i}{\sqrt{2}}|1\rangle$$

We can represent this state using ket notation as:

$$|\psi\rangle = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{i}{\sqrt{2}} \end{pmatrix}$$

To find the corresponding bra vector, we take the conjugate transpose of the ket vector:

$$\langle\psi| = \begin{pmatrix} \frac{1}{\sqrt{2}}^* & \frac{i}{\sqrt{2}}^* \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{i}{\sqrt{2}} \end{pmatrix}$$

These bra and ket vectors can be used in various quantum mechanical operations and calculations, such as calculating inner products, expectation values, and probabilities.